PREP 4, LESSONS 7–8

Combining resistors and I–V graphs

Resistors in series and parallel (required practical); I–V characteristics

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons, 1 mark each

1 Describe how to calculate the total resistance of two resistors in series. [1]

2 Describe the potential difference across components in parallel. [1]

3 Describe how the length of a wire affects its resistance. [1]

Key ideas LESSONS 7–8

Show your working

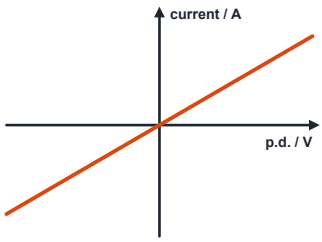
4 Two 10 Ω resistors are connected in parallel.

a Their total resistance is | less than 10 Ω | 10 Ω | 20 Ω | [1]

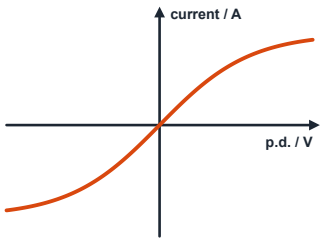
b Explain why adding a resistor in parallel decreases the total resistance. [2]

5 These are the I–V characteristics of three components.

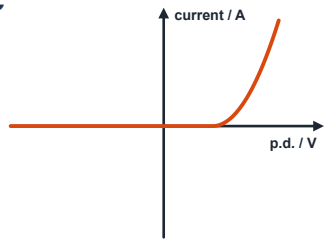
A



B



C



a Which graph shows a filament lamp? | A | B | C | [1]

b Which component lets current flow in one direction only? | resistor | filament lamp | diode | [1]

c Explain why the resistance of a filament lamp increases as the current increases. [2]

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